Proceedings of the Combustion Institute 41 (2025) 105799

Contents lists available at ScienceDirect

Proceedings of the Combustion Institute

journal homepage: www.elsevier.com/locate/proci

A regime diagram for detonation–turbulence interactions

Benjamin Breer *, Guillaume Blanquart

Department of Mechanical & Civil Engineering, California Institute of Technology, Pasadena, CA 91125, USA

ARTICLE INFO

Keywords:
Detonation
Turbulence
Regime diagram

ABSTRACT

The propagation of a detonation wave in a laminar fluid and the intrinsic instabilities leading to a 3-D cellular structure has been established in the literature. In contrast, the behavior of detonation propagation in turbulent flows lacks a general characterization. This study develops a theoretical framework for probing this interaction. Similar to the Borghi–Peters regime diagram for turbulent premixed flames, a detonation regime diagram is proposed which relates qualitatively the turbulence intensity (root-mean-square velocity fluctuation normalized by the Chapman–Jouguet detonation propagation speed) to the integral length scale normalized by the induction length. By comparing the length and time scales of a detonation to turbulence, four different regimes of detonation wave structure are identified: (I) the laminar detonation regime, wherein most of the detonation literature exists; (II) the thin detonation regime, in which the interior structure of the detonation is smaller than, and therefore unperturbed by, pre-shock turbulence; (III) the wrinkled reaction zone regime, where ignition occurs much faster than the Kolmogorov time scale; and (IV) the shortened induction zone regime, where the Kolmogorov time scale is shorter than the ignition delay time. The vertical position of the regime boundaries is controlled by a dimensionless "detonation constant" and is dependent on the thermochemical state of the unburnt mixture. Prior direct numerical simulations are located in regimes III and IV and substantiate a practical boundary between these regimes.

1. Introduction

Detonations are supersonic combustion waves that exist in nature (e.g., supernovae) and terrestrial energy conversion processes (e.g., detonation engines, accidental explosions). The theoretical structure of a one-dimensional (1-D) Chapman–Jouguet (CJ) detonation is commonly referred to as the Zel'dovich–von Neumann–Döring (ZND) model [1–3] and consists of three components: the lead shock, induction zone, and reaction zone. The characteristic length and velocity scale of a detonation are the induction length ($\mathcal{L}$) and CJ speed (U_{CJ}), respectively. Both quantities are determined by the composition and thermodynamic state of the unburnt gaseous mixture.

The hydrodynamic structure and inherent cellular instability (cell sizes of $\sim 10-100\mathcal{L}$) of multi-dimensional detonation waves propagating in laminar flows has been the subject of numerous studies (e.g., [4–8]). Unfortunately, in most practical applications such as advanced propulsion systems (e.g., rotating detonation engines, or RDEs), the unburnt flow is turbulent. Seminal reviews (e.g., [7–9]) have considered turbulence as a driving mechanism behind non-ideal detonation behavior. Early work on 3-D detonation–turbulence interaction used Direct Numerical Simulation (DNS) but oversimplified the detonation structure by assuming an infinitely thin induction zone [10,11]. The authors demonstrated that omission of the induction length largely reduced the detonation–turbulence interaction to a shock–turbulence interaction (which has been studied extensively, e.g., [12–16]). Massa and coworkers [17] accounted for the induction length to determine its influence. They performed two DNS with one-step chemistry (low and high heat release cases) and suggested that the ratio of the induction length to the Taylor micro scale (fixed at unity) could control the detonation–turbulence interaction. Using a similar simulation framework, Jin et al. [18] performed four simulations with different mean flow and turbulent Mach numbers. The reaction zone in the cases with turbulent inflow possessed small-scale "corrugations." Maxwell et al. [19] conducted mostly 2-D and one 3-D Large Eddy Simulation (LES) with one-step chemistry and varied the turbulence by tuning the eddy viscosity and dissipation rate models. They showed that higher turbulence intensity led to longer ignition delay times and formation of unburnt gas pockets in the wake. However, turbulence was imposed as random perturbations in the pre-shock density. Recently, Iwata et al. [20] and Suzuki et al. [21] performed DNS with detailed chemistry of hydrogen/oxygen detonations interacting with isotropic turbulence. Both studies reported a qualitative "wrinkling" of the reaction front, while Suzuki et al. [21] found that the induction zone shortened in their simulation with the highest turbulence intensity.

* Corresponding author.
E-mail address: bbreer@caltech.edu (B. Breer).

https://doi.org/10.1016/j.proci.2025.105799
Received 21 April 2025; Accepted 24 July 2025
Available online 9 August 2025
1540-7489/© 2025 The Combustion Institute. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

A common feature across all of the literature is the lack of a theoretical framework for predicting and describing the detonation–turbulence interaction for a wide range of scales and structures. Interestingly, detonation waves share several of the same features present in deflagration; both are characterized by a length and velocity scale. In studying turbulent premixed flames, Borghi [22], Peters [23], and others considered the controlling parameters of the flame-turbulence interaction to be (1) the ratio of the turbulent integral length scale (largest eddy size) to the laminar flame thickness, l/l_F; and (2) the turbulence intensity, i.e., the ratio of the root-mean-square (r.m.s.) velocity to the laminar flame speed, u'/S_L. The result was a regime diagram that identified and parameterized different flame structures using $l/l_F - u'/S_L$ coordinates. Inspired by the turbulent premixed combustion regime diagram, the objective of the present work is to develop a "detonation regime" diagram. A separate regime diagram is necessary, as the combustion mechanisms of detonation and premixed flames are different: detonation involves ignition through adiabatic shock compression, whereas deflagration is a diffusion-dominant process. The relevant parameters of the detonation–turbulence interaction are (1) the ratio of the integral length scale to the induction length, $l/\mathcal{L}$; and (2) the ratio of the r.m.s. velocity to the CJ speed, u'/U_{CJ}.

The remainder of this paper is organized as follows. First, the detonation regime diagram is developed in Section 2. In Section 3, the sensitivity of the regime boundaries to the shock–turbulence interaction and different unburnt gas conditions is assessed. Finally, in Section 4, prior DNS studies are located on the regime diagram, and the practicality of the regime boundaries is discussed.

2. Detonation regime diagram

Fig. 1 shows a representative detonation regime diagram on a log–log scale for a mixture of $2H_2$-O_2-17Ar at 20 kPa and 295 K. The Shock and Detonation Toolbox [24] was used to compute the detonation quantities. The vertical axis is the turbulence intensity, u'/U_{CJ}; the horizontal axis is the integral length scale normalized by the induction length, $l/\mathcal{L}$. The induction length is defined as the distance from the shock front to the beginning of the reaction zone where the temperature gradient achieves a maximum [25]. In the following subsections, we derive the boundaries between regimes. Quantities denoted with a subscript "u" and "N" refer to the unburnt and von Neumann state, respectively.

2.1. Fundamental turbulence relations

We begin by introducing the fundamental scales and relations of turbulence; some relationships assume homogeneous isotropic turbulence (HIT) [26]. The interaction of turbulence with the lead shock introduces anisotropy and is addressed in Section 3.1.

The integral length scale is defined as the ratio of the r.m.s. velocity (u') and energy dissipation rate (ϵ):

$$l \equiv \frac{(u')^3}{\epsilon}\,. \tag{1}$$

The Kolmogorov length scale – which defines the size of the smallest eddies – is strictly a function of the kinematic viscosity (ν) and energy dissipation rate:

$$\eta \equiv \left(\frac{\nu^3}{\epsilon}\right)^{1/4}\,. \tag{2}$$

The time scale associated with the smallest length scale is the Kolmogorov time scale and is defined as:

$$\tau_\eta \equiv \left(\frac{\nu}{\epsilon}\right)^{1/2}\,. \tag{3}$$

The eddy turnover time, denoted τ_l, characterizes the large scales of the turbulent flow and is defined as:

$$\tau_l \equiv \frac{l}{u'}\,. \tag{4}$$

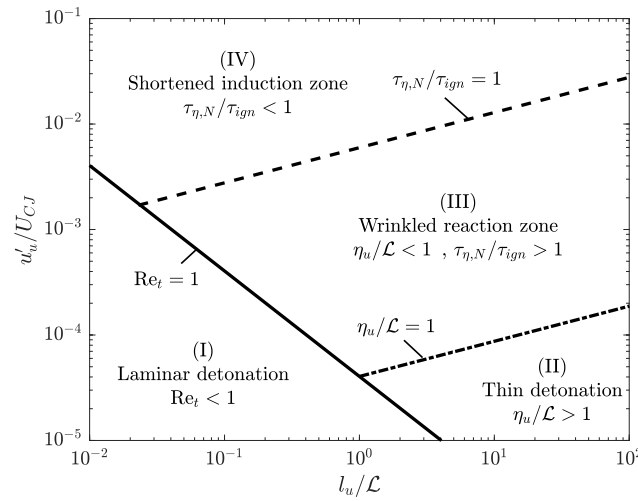

Fig. 1. Detonation regime diagram for $2H_2$-O_2-17Ar at $P_u = 20$ kPa and $T_u = 295$ K.

It can be shown that the ratio of the eddy turnover time to the Kolmogorov time scale is

$$\frac{\tau_l}{\tau_\eta} = \mathrm{Re}_t^{1/2}\,, \tag{5}$$

where $\mathrm{Re}_t \equiv u'\,l/\nu$ is the turbulent Reynolds number. Using the above definitions, the ratio of the smallest to largest length scales can be expressed as

$$\frac{\eta}{l} = \mathrm{Re}_t^{-3/4}\,. \tag{6}$$

Finally, the turbulent Mach number is the ratio of the r.m.s. velocity to the sound speed, i.e., $M_t = u'/a$.

2.2. The laminar detonation regime

The regime labeled I in Fig. 1 corresponds to laminar unburnt fluids where $\mathrm{Re}_t < 1$. The bulk of gaseous detonation literature is confined to this "locally laminar" detonation regime.

The turbulent Reynolds number can be written in terms of the controlling parameters $l/\mathcal{L}$ and u'/U_{CJ}:

$$\mathrm{Re}_t = \frac{u'_u\,l_u}{\nu_u} = \left(\frac{U_{CJ}\mathcal{L}}{\nu_u}\right)\left(\frac{u'_u}{U_{CJ}}\right)\left(\frac{l_u}{\mathcal{L}}\right)\,. \tag{7}$$

The term $U_{CJ}\mathcal{L}/\nu_u$, hereby denoted as β_{CJ}, is a dimensionless constant determined by the chemical composition and thermodynamic state of a gaseous mixture undergoing detonation.

$$\beta_{CJ} = U_{CJ}\mathcal{L}/\nu_u \tag{8}$$

Eq. (7) can be rearranged to identify the "laminar-turbulent" dividing line ($\mathrm{Re}_t = 1$):

$$\frac{u'_u}{U_{CJ}} = \left(\frac{l_u}{\mathcal{L}}\right)^{-1}\beta_{CJ}^{-1}\,. \tag{9}$$

Regime I corresponds to $u'_u/U_{CJ} < \left(l_u/\mathcal{L}\right)^{-1}\beta_{CJ}^{-1}$. Lines of constant Re_t have a slope of -1 on the detonation regime diagram.

2.3. Turbulent detonations: length vs. time scales

In turbulent premixed combustion, the Karlovitz number (Ka) is defined as the ratio of the flame time scale (τ_F) to the Kolmogorov time scale, or equivalently as the squared ratio of the flame thickness (l_F) to the Kolmogorov length scale [23]:

$$\mathrm{Ka} \equiv \frac{l_F^2}{\eta^2} = \frac{\tau_F}{\tau_\eta}\,. \tag{10}$$

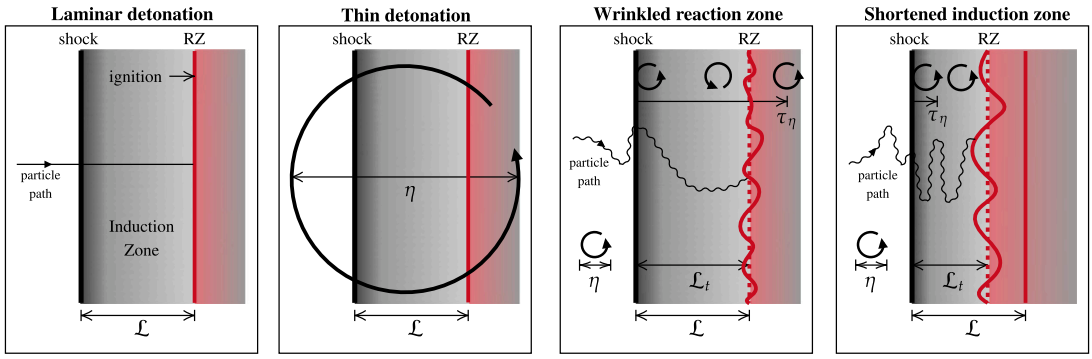

Fig. 2. Schematic representation of detonation–turbulence interactions for each regime. "RZ" refers to the reaction zone front, and $\mathcal{L}_t$ is the effective induction length for a turbulent detonation.

Hence, a Karlovitz number of one corresponds to both $l_F = \eta$ and $\tau_F = \tau_\eta$ and defines a single regime boundary. In contrast, for detonation–turbulence interaction, unity length and time scale ratios do not result in the same regime boundary due to the density discontinuity at the shock.

2.3.1. Length scale ratio

Using Eq. (6), the Kolmogorov-induction length ratio ($\eta/\mathcal{L}$) is expressed as:

$$\frac{\eta_u}{\mathcal{L}} = \frac{l_u}{\mathcal{L}}\frac{\eta_u}{l_u} = \frac{l_u}{\mathcal{L}}\mathrm{Re}_t^{-3/4}. \tag{11}$$

Substituting Eq. (7) into Eq. (11), the ratio becomes:

$$\frac{\eta_u}{\mathcal{L}} = \left(\frac{l_u}{\mathcal{L}}\right)^{1/4}\left(\frac{u'_u}{U_{CJ}}\right)^{-3/4}\beta_{CJ}^{-3/4}. \tag{12}$$

The regime boundary corresponding to a unity Kolmogorov-induction length ratio ($\eta_u/\mathcal{L} = 1$) is indicated by the dashed-dotted line in Fig. 1 and is given by the following equation:

$$\frac{u'_u}{U_{CJ}} = \left(\frac{l_u}{\mathcal{L}}\right)^{1/3}\beta_{CJ}^{-1}. \tag{13}$$

This regime boundary has a slope of 1/3 on the regime diagram.

2.3.2. Time scale ratio

The time scale of a detonation is the ignition delay time (τ_{ign}),

$$\tau_{ign} = \mathcal{L}/U_N, \tag{14}$$

where U_N is the von Neumann speed (immediately post-shock). The relevant turbulence time scale for comparison with τ_{ign} is the von Neumann Kolmogorov time scale, $\tau_{\eta,N}$. The associated time scale ratio is expressed as:

$$\frac{\tau_{\eta,N}}{\tau_{ign}} = \frac{\tau_{\eta,N}}{\tau_{\eta,u}}\frac{\tau_{\eta,u}}{\tau_{ign}}. \tag{15}$$

Using Eq. (5) to obtain $\tau_{\eta,u}$ in terms of $\tau_{l,u}$ and Re_t, Eq. (15) becomes:

$$\frac{\tau_{\eta,N}}{\tau_{ign}} = \frac{\tau_{\eta,N}}{\tau_{\eta,u}}\frac{\tau_{l,u}}{\tau_{ign}}\mathrm{Re}_t^{-1/2}. \tag{16}$$

Then, substituting Eq. (4) for $\tau_{l,u}$, Eq. (7) for Re_t, and Eq. (14) for τ_{ign} yields:

$$\frac{\tau_{\eta,N}}{\tau_{ign}} = \left(\frac{l_u}{\mathcal{L}}\right)^{1/2}\left(\frac{u'_u}{U_{CJ}}\right)^{-3/2}\beta_{CJ}^{-1/2}\frac{U_N}{U_{CJ}}\frac{\tau_{\eta,N}}{\tau_{\eta,u}}. \tag{17}$$

The regime boundary corresponding to a unity time scale ratio ($\tau_{\eta,N}/\tau_{ign} = 1$) is given by:

$$\frac{u'_u}{U_{CJ}} = \left(\frac{l_u}{\mathcal{L}}\right)^{1/3}\beta_{CJ}^{-1/3}\left(\frac{U_N}{U_{CJ}}\frac{\tau_{\eta,N}}{\tau_{\eta,u}}\right)^{2/3}. \tag{18}$$

This dividing line (dashed line in Fig. 1) has the same slope as the $\eta_u/\mathcal{L} = 1$ boundary given by Eq. (13) but contains different constants resulting in separate boundaries for unity length and time scale ratios. The velocity ratio (U_N/U_{CJ}) stems from the density jump across the shock, whereas the Kolmogorov time scale ratio ($\tau_{\eta,N}/\tau_{\eta,u}$) is due to the shock–turbulence interaction (addressed in Section 3.1).

2.4. The thin detonation regime

When $\eta_u/\mathcal{L} > 1$, the smallest turbulent eddies are larger than the induction length, so the detonation is "thin" compared to all scales of the incoming turbulence. In this regime, the entire induction zone is embedded within the size of the smallest eddies as shown in Fig. 2. Since the flow is quasi-laminar inside eddies of size η_u, the internal structure of the detonation remains unperturbed. This regime II corresponds to:

$$\frac{u'_u}{U_{CJ}} < \left(\frac{l_u}{\mathcal{L}}\right)^{1/3}\beta_{CJ}^{-1}. \tag{19}$$

The leading shock front is still expected to exhibit curvature but at scales larger than the induction length.

2.5. The wrinkled reaction zone regime

Regime III corresponds to the region where both $\eta_u/\mathcal{L} < 1$ and $\tau_{\eta,N}/\tau_{ign} > 1$ hold. In this regime, turbulent fluctuations are inside the induction zone, but the turnover time for the smallest eddies of size η_N is longer than the ignition delay time (see Fig. 2). Thus, the reaction zone is expected to be "wrinkled", and yet the effective induction length should remain close to the CJ-predicted induction length ($\mathcal{L}$). The regime is delimited by

$$\frac{u'_u}{U_{CJ}} > \left(\frac{l_u}{\mathcal{L}}\right)^{1/3}\beta_{CJ}^{-1} \tag{20}$$

and

$$\frac{u'_u}{U_{CJ}} < \left(\frac{l_u}{\mathcal{L}}\right)^{1/3}\beta_{CJ}^{-1/3}\left(\frac{U_N}{U_{CJ}}\frac{\tau_{\eta,N}}{\tau_{\eta,u}}\right)^{2/3}. \tag{21}$$

2.6. The shortened induction zone regime

Finally, when $\tau_{\eta,N}/\tau_{ign} < 1$, the time scale of the smallest eddies of size η_N is shorter than the ignition delay time of the mixture. In this regime, fluid particles take a tortuous path through the induction zone due to large turbulent fluctuations before ignition occurs. Thus, ignition occurs at random distances from the shock front that, on average, are shorter than the CJ-predicted induction length. This is represented schematically in Fig. 2, where $\mathcal{L}_t$ is the effective induction length for a turbulent detonation. This regime IV is labeled the "shortened induction zone" regime and occupies the upper left portion of Fig. 1.

Theoretically, the regime is delimited by

$$\frac{u'_u}{U_{CJ}} > \left(\frac{l_u}{\mathcal{L}}\right)^{1/3}\beta_{CJ}^{-1/3}\left(\frac{U_N}{U_{CJ}}\frac{\tau_{\eta,N}}{\tau_{\eta,u}}\right)^{2/3}. \tag{22}$$

Practically, a time scale ratio much smaller than unity, $\tau_{\eta,N}/\tau_{ign} \ll 1$, would be necessary to induce a shortened induction zone. A more practical time scale regime boundary will be substantiated when reviewing past numerical studies in Section 4.

2.7. Role of detonation cell width

The cellular instability length scale or cell width (λ_{cell}) is related to the induction length as $\lambda_{\text{cell}} = A\mathcal{L}$. This relationship holds for a wide range of mixtures, and the proportionality constant A ranges from 10 to 100 [5]. For the conditions in Fig. 1, the experimentally measured cell width is $\lambda_{\text{cell}} \approx 70\mathcal{L}$ [5]. In the situation where the integral length scale is smaller than the cell width, i.e., $l_u < \lambda_{\text{cell}}$, all the turbulence resides within a single cell. Thus, the turbulence interacts with a locally planar detonation, and the cell width does not govern detonation–turbulence interaction. This region corresponds to $l_u/\lambda_{\text{cell}} < 1$ or equivalently for Fig. 1, $l_u/\mathcal{L} < A \approx 70$. The description of regimes corresponding to $l_u/\lambda_{\text{cell}} > 1$ is beyond the scope of the present work.

3. Detonation regime boundary constants

In this section, we discuss first two aspects which were omitted in deriving the previous regime diagram: (1) the shock–turbulence interaction, and (2) the "detonation constant" (β_{CJ}) which controls the vertical position of the regime boundaries. Then, we consider the spatial variation of viscosity.

3.1. Shock–turbulence interaction

Before interacting with the reaction zone of a detonation, turbulence crosses the leading shock. Shock–turbulence interaction has been the focus of numerous studies (e.g., [12–16].) It is not within the scope of the present study to provide an exhaustive review. Only the main observations that relate to the present analysis are mentioned.

The regime boundary given by Eq. (18) includes the ratio $\tau_{\eta,N}/\tau_{\eta,u}$ which is the result of the shock–turbulence interaction. There are primarily two ways in which a shock affects turbulence: (1) vorticity is amplified while turbulent length scales decrease; and (2) the turbulence becomes anisotropic immediately post-shock [15,16]. Vorticity (ω) depends strongly on the smallest scales; its magnitude scales with the Kolmogorov time scale:

$$\omega \sim \frac{u(\eta)}{\eta} \sim \tau_\eta^{-1} . \tag{23}$$

Mahesh et al. [12] showed using linear interaction analysis and the Rankine–Hugoniot relations that the vorticity jump across a shock scales as the mean density ratio. Larsson and Lele [15] demonstrated through comparison with DNS data that this vorticity scaling was satisfactory for a wide range of incident Mach number $M \in [1.05, 6]$. Thus, for isotropic turbulence interacting with a detonation, the vorticity across the lead shock scales reasonably as:

$$\frac{\omega_N}{\omega_u} \sim \frac{\rho_N}{\rho_u} . \tag{24}$$

Combining Eqs. (23) and (24), the jump in Kolmogorov time scale across the lead shock is $\tau_{\eta,N}/\tau_{\eta,u} \sim \rho_u/\rho_N$. In addition, the Rankine–Hugoniot relation for the velocity jump is $U_N/U_{CJ} = \rho_u/\rho_N$. The time scale ratio given by Eq. (17) becomes:

$$\frac{\tau_{\eta,N}}{\tau_{\eta,ign}} = \left(\frac{l_u}{\mathcal{L}}\right)^{1/2} \left(\frac{u'_u}{U_{CJ}}\right)^{-3/2} \beta_{CJ}^{-1/2} \left(\frac{\rho_u}{\rho_N}\right)^2 . \tag{25}$$

Hence, the unity time scale regime boundary given by Eq. (18) reduces to:

$$\frac{u'_u}{U_{CJ}} = \left(\frac{l_u}{\mathcal{L}}\right)^{1/3} \beta_{CJ}^{-1/3} \left(\frac{\rho_u}{\rho_N}\right)^{4/3} . \tag{26}$$

While turbulence becomes anisotropic immediately behind the shock, Larsson et al. [16] showed that turbulence returns to isotropy after about 10 convected Kolmogorov time scales. In a detonation, this anisotropy relaxation distance is $10\, U_N\, \tau_{\eta,N}$. Hence, for $\tau_{\eta,N}/\tau_{ign} < 0.1$ in the regime diagram (Fig. 1), post-shock turbulence becomes isotropic again before interacting with the reaction zone. Conversely, in regimes where $\tau_{\eta,N}/\tau_{ign} > 0.1$, post-shock turbulence is still anisotropic when it interacts with the reaction zone, but the regime boundaries do not change.

3.2. Impact of unburnt gas conditions

The regime boundaries, given by Eqs. (9), (13), and (26), are functions of β_{CJ} and the density ratio $(\rho_u/\rho_N)^{4/3}$. These two quantities fix the vertical position of the regime boundaries. Unburnt gas conditions that could change their values are the pressure (P_u), temperature (T_u), diluent mole fraction (X_{Ar} or X_{N_2}), and equivalence ratio (ϕ). Fig. 3 shows the sensitivity of β_{CJ}^{-1} to changing unburnt gas conditions for a hydrogen/oxygen/argon mixture. The density ratio parameter, $(\rho_u/\rho_N)^{4/3}$, varies weakly from 0.122 to 0.223 across all unburnt gas conditions tested and will not be discussed further.

Fig. 3(a) shows that the regime boundary constant β_{CJ}^{-1} remains approximately constant when the unburnt pressure varies from 20 to 200 kPa with all other unburnt gas conditions fixed. The induction length is inversely proportional to unburnt pressure, as is the kinematic viscosity for an ideal gas, whereas the CJ speed is invariant to pressure changes. Thus, the net effect is that β_{CJ}^{-1} is insensitive to changes in unburnt pressure. For the remainder of the parametric study, the unburnt pressure is fixed at 20 kPa.

In Fig. 3(b), β_{CJ}^{-1} grows proportional to $T_u^{3/2}$ as the unburnt temperature varies from 300 to 800 K with all other parameters fixed. This trend is driven by the kinematic viscosity which varies as $\nu_u \propto T_u^{1.5}$, while the induction length and CJ speed are invariant to changes in unburnt temperature.

Finally, in Fig. 3(c), the equivalence ratio is varied from lean to rich conditions for different argon dilution levels with all other parameters fixed. For dilution levels of 60% or less, β_{CJ}^{-1} remains roughly constant for $0.5 \le \phi \le 1.5$. For fuel rich conditions ($\phi > 1.5$), the detonation constant decreases, with the effect amplified for dilution levels higher than 60%.

In Fig. 3(a–c), there is no significant change in β_{CJ}^{-1} for argon dilution levels below 60%. For greater dilutions, β_{CJ}^{-1} decreases primarily because of the kinetic role that diluent has in elongating the induction zone.

3.3. Validity of using unburnt kinematic viscosity

As turbulence crosses a detonation, it experiences a spatially varying viscosity. The jumps in temperature and pressure across the shock counteract each other and result in a relatively small change in viscosity ($\nu \sim T^{1.5}P^{-1}$). For the unburnt gas conditions in Fig. 1, the viscosity changes by only 5% across the three states as shown in Fig. 4: the unburnt state, the von Neumann state, and the reaction zone defined by the position of maximum temperature gradient.

For the parameter space investigated in Section 3.2, the maximum change in viscosity between the unburnt, von Neumann, and reaction zone states was only 40%. Thus, the regime diagram is largely insensitive to which state the viscosity is extracted from and hence the unburnt kinematic viscosity is used.

4. Probing the regime diagram

We now locate all relevant detonation–turbulence interaction literature on the regime diagram. We restrict ourselves to studies that performed direct numerical simulations. We do not locate laminar detonation studies corresponding to regime I. The studies [17,20,21] are reviewed below and placed on two normalized regime diagrams (Fig. 5).

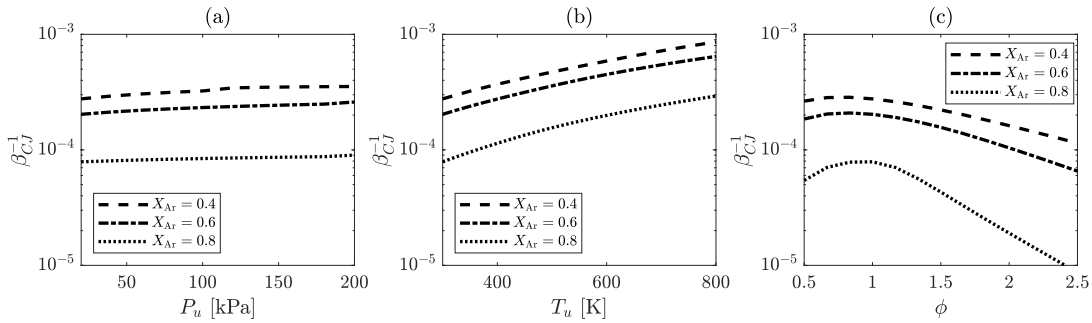

Fig. 3. Sensitivity of the detonation regime diagram boundary constant (β_{CJ}^{-1}) to changes in unburnt gas conditions. (a): varying unburnt pressure at $T_u = 300\,\mathrm{K}$ and $\phi = 1$. (b): varying unburnt temperature at $P_u = 20\,\mathrm{kPa}$ and $\phi = 1$. (c) varying equivalence ratio at $P_u = 20\,\mathrm{kPa}$ and $T_u = 300\,\mathrm{K}$. For all cases (a)–(c), three different argon dilutions (X_{Ar}) are presented.

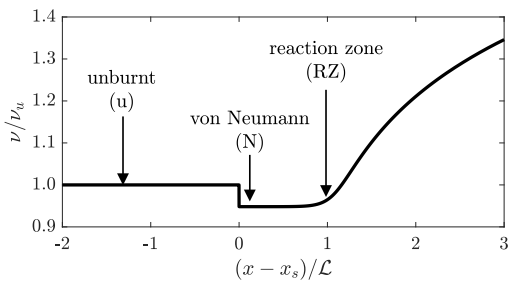

Fig. 4. Kinematic viscosity (normalized by the unburnt value) as a function of position from the lead shock for the same conditions as in Fig. 1.

Table 1
Estimates of $\tau_{\eta,N}/\tau_{ign}$ and practical regime for DNS of detonation–turbulence interaction.

Reference	Case	$\tau_{\eta,N}/\tau_{ign}$	Regime
Jackson et al. [10,11]		∞	II
Massa et al. [17]	LHR	0.0260	IV
–	HHR	0.0358	III
Iwata et al. [20]	1-T	0.271	III
–	2-T	0.418	III
Suzuki et al. [21]	T1	0.164	III
–	T2	0.0484	III
–	T3	0.0225	IV

4.1. Prior direct numerical simulations

Massa et al. [17] performed two simulations with one-step chemistry at a specific heat ratio of $\gamma = 1.2$ and turbulent Mach number of $M_t = 0.235$. The two cases were an inflow Mach number of $M = 4.0$ and $M = 5.5$ corresponding to a low heat release (LHR) and high heat release (HHR) case, respectively. The numerical schlieren fields for both cases revealed a wrinkled reaction zone. In the LHR case, the induction zone shortened by roughly 16% consistent with the expected behavior in regime IV; in the HHR case, the induction zone shortened by only 3.8% and corresponds to the boundary between regimes III and IV.

Using a similar simulation framework as Massa et al., Jin et al. [18] performed DNS of four cases of detonation–turbulence interaction. Because the kinematic viscosity and Reynolds number at the inlet to the detonation domain were not reported, these cases could not be located on the regime diagram. However, the reported qualitative findings support the description of the wrinkled reaction zone regime.

Iwata et al. [20] performed simulations of stoichiometric hydrogen/oxygen/argon mixtures at two different dilution levels and unburnt pressures. The first simulation was of a $2H_2$-O_2-7Ar mixture at an unburnt pressure of 60 kPa (indicated by the red diamond and label "1-T" in Fig. 5); the second simulation was of a $2H_2$-O_2-Ar mixture at 40 kPa (indicated by the red triangle and label "2-T"). Both simulations were reported to have the same turbulent Mach number and Reynolds number and thus fall on a line with slope of −1. The reaction fronts, defined as isosurfaces of half hydrogen mass fraction, were found to exhibit small-scale "corrugations." In addition, the probability density function of induction length (for either simulation) did not deviate significantly from the corresponding laminar case. The previous observations are consistent with the expected behavior in the wrinkled reaction zone regime.

Suzuki et al. [21] considered a stoichiometric hydrogen/oxygen mixture (without dilution) at a fixed unburnt pressure of 40 kPa and varied the turbulence intensity. In simulations T1 and T2, which had lower turbulence intensities, the distributions of induction length did not change significantly compared to the laminar cases. The reaction fronts, defined in the same manner as Iwata et al. [20], appeared wrinkled for cases T1 and T2. The third simulation, labeled T3, had the highest turbulence intensity, and the induction length distribution was reported to shift toward shorter induction lengths consistent with regime IV.

Jackson and coworkers [10,11] treated the detonation as a discontinuity. Thus, for any pre-shock turbulence intensity, their DNS would reside within the thin detonation regime. The authors only investigated the transformation of turbulence across an infinitely thin detonation and hence could not discuss changes to the internal structure of the detonation.

4.2. A more practical time scale regime boundary

In order to classify quantitatively which regimes the DNS reside in, estimates for their respective time scale ratios $\tau_{\eta,N}/\tau_{ign}$ are necessary. Using Eqs. (15), (23), and (24), the ratio of the von Neumann Kolmogorov time scale to the ignition delay time can be expressed as:

$$\frac{\tau_{\eta,N}}{\tau_{ign}} = \frac{\rho_u}{\rho_N}\frac{\tau_{\eta,u}}{\tau_{ign}} . \tag{27}$$

Then, $\tau_{\eta,u}$ is determined by rearranging Eq. (2) for ϵ and substituting into Eq. (3):

$$\tau_{\eta,u} = \frac{\eta_u^2}{\nu_u} . \tag{28}$$

Finally, substituting Eq. (28) into Eq. (27) yields:

$$\frac{\tau_{\eta,N}}{\tau_{ign}} = \frac{\rho_u}{\rho_N}\frac{\eta_u^2}{\nu_u}\frac{U_N}{\mathcal{L}} . \tag{29}$$

For Suzuki et al. [21], η_u was computed from Eq. (6) using the reported unburnt integral length scale and r.m.s. velocity. Based on the estimated time scale ratio listed in Table 1, a more practical time scale ratio regime boundary is around $\tau_{\eta,N}/\tau_{ign} = 0.03$. Fig. 5 shows this modified time scale regime boundary.

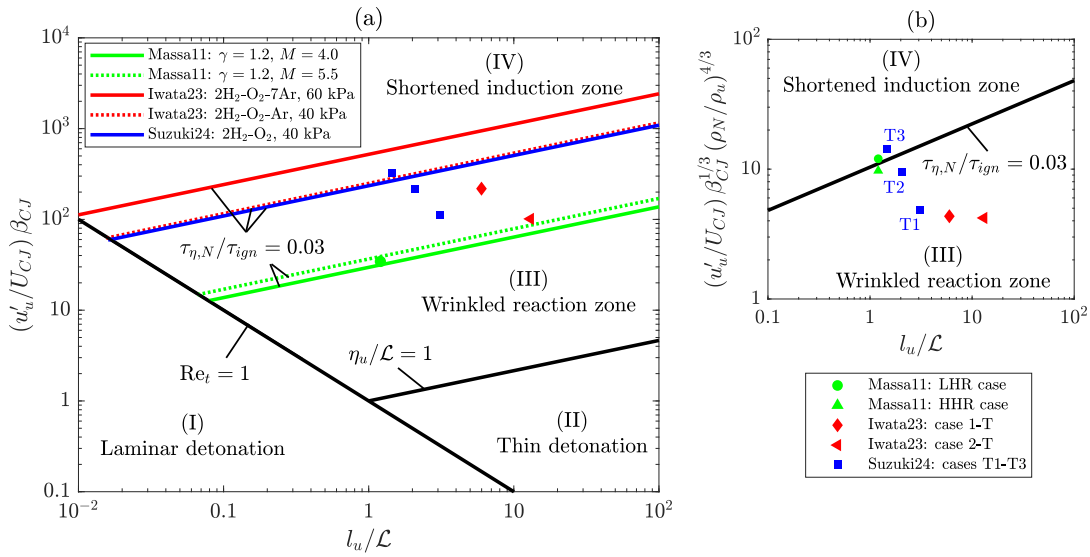

Fig. 5. Regime boundaries (lines) and simulations (symbols) from three DNS studies [17,20,21] located on two normalized regime diagrams. (a) normalization of the vertical axis by β_{CJ} collapses the $\mathrm{Re}_t = 1$ and $\eta_u/\mathcal{L} = 1$ regime boundaries. (b) normalization of the vertical axis by $\beta_{CJ}^{1/3}\ (\rho_N/\rho_u)^{4/3}$ collapses the $\tau_{\eta,N}/\tau_{ign} = 0.03$ regime boundary.

5. Conclusions

A detonation regime diagram was developed through qualitative comparisons of the length and time scales of detonation to turbulence. Three regime boundaries were derived: (1) unity turbulent Reynolds number, $\mathrm{Re}_t = 1$, (2) unity length scale ratio, $\eta_u/\mathcal{L} = 1$, and (3) unity time scale ratio, $\tau_{\eta,N}/\tau_{ign} = 1$. These boundaries yielded four different regimes corresponding to unique detonation structure. All three regime boundaries are a function of a dimensionless constant, $U_{CJ}\mathcal{L}/\nu_u$, which is only a function of the unburnt gas composition and thermodynamic state. The unburnt pressure has no impact on this parameter, whereas unburnt temperature, equivalence ratio, and dilution level have a limited effect.

All prior DNS of detonation–turbulence interaction were located on the proposed detonation regime diagram. The time scale ratio $\tau_{\eta,N}/\tau_{ign}$ was computed for each prior DNS and revealed a more practical time scale ratio regime boundary to be $\tau_{\eta,N}/\tau_{ign} = 0.03$. The regime diagram could be a valuable tool for characterizing the physics of detonation engines.

Novelty and significance statement

This work proposes, to the authors' knowledge, the first-ever regime diagram for describing the detonation–turbulence interaction (inspired by the regime diagram for turbulent premixed flames). The regime diagram compartmentalizes the detonation–turbulence interaction into four distinct regimes that are substantiated by prior DNS studies. There are two major applications of the proposed regime diagram: (1) characterizing the physics of rotating detonation engines (RDEs), a major research thrust in advanced propulsion; and (2) guiding the design of future experimental and numerical studies on detonation–turbulence interaction.

CRediT authorship contribution statement

Benjamin Breer: Writing – original draft, Funding acquisition, Formal analysis, Visualization, Conceptualization. **Guillaume Blanquart:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Benjamin Breer was supported by the Department of Defense (DoD), United States through the National Defense Science & Engineering Graduate (NDSEG) Fellowship Program, United States 32 CFR part 168a, awarded by the Air Force Research Laboratory (AFRL).

References

[1] Y. Zel'dovich, On the theory of the propagation of detonation in gaseous systems, J. Exp. Theor. Phys. 10 (1940) 542–568, available in translation as NACA TM 1261 (1950).
[2] J. von Neumann, Theory of detonation waves, Collect. Work. 6 (1942) 203–218.
[3] W. Döring, On detonation processes in gases, Ann. Phys. 435 (6–7) (1943) 421–436.
[4] W. Fickett, W. Davis, Detonation: Theory and Experiment, Dover Publications, Mineola, NY, 2000.
[5] J.M.K. Austin, The Role of Instability in Gaseous Detonation (Ph.D. thesis), California Institute of Technology, Pasadena, CA, USA, 2003.
[6] M.I. Radulescu, G.J. Sharpe, C.K. Law, J.H.S. Lee, The hydrodynamic structure of unstable cellular detonations, J. Fluid Mech. 580 (2007) 31–81.
[7] J. Lee, The Detonation Phenomenon, Cambridge University Press, Cambridge, UK, 2008.
[8] J. Shepherd, Detonation in gases, Proc. Combust. Inst. 32 (2009) 83–98.
[9] V. Raman, S. Prakash, M. Gamba, Nonidealities in rotating detonation engines, Annu. Rev. Fluid Mech. 55 (2023) 639–674.
[10] T.L. Jackson, A.K. Kapila, M.Y. Hussaini, Convection of a pattern of vorticity through a reacting shock wave, Phys. Fluids A- Fluid 2 (1990) 1260–1268.
[11] T.L. Jackson, M.Y. Hussaini, H.S. Ribner, Interaction of turbulence with a detonation wave, Phys. Fluids A- Fluid 5 (1993) 745–749.
[12] K. Mahesh, S.K. Lele, P. Moin, The influence of entropy fluctuations on the interaction of turbulence with a shock wave, J. Fluid Mech. 334 (1997) 353–379.
[13] S. Lee, S.K. Lele, P. Moin, Interaction of isotropic turbulence with shock waves: effect of shock strength, J. Fluid Mech. 340 (1997) 225–247.
[14] Y. Andreopoulos, J.H. Agui, G. Briassulis, Shock wave–turbulence interactions, Annu. Rev. Fluid Mech. 32 (2000) 309–345.
[15] J. Larsson, S.K. Lele, Direct numerical simulation of canonical shock/turbulence interaction, Phys. Fluids 21 (2009) 126101.
[16] J. Larsson, I. Bermejo-Moreno, S.K. Lele, Reynolds- and mach-number effects in canonical shock–turbulence interaction, J. Fluid Mech. 717 (2013) 293–321.
[17] L. Massa, M. Chauhan, F. Lu, Detonation–turbulence interaction, Combust. Flame 158 (2011) 1788–1806.

[18] T. Jin, K. Luo, Q. Dai, J. Fan, Simulations of cellular detonation interaction with turbulent flows, AIAA J. 54 (2016) 419–433.
[19] B.M. Maxwell, R.R. Bhattacharjee, S.S.M. Lau-Chapdelaine, S.A.E.G. Falle, G.J. Sharpe, M.I. Radulescu, Influence of turbulent fluctuations on detonation propagation, J. Fluid Mech. 818 (2017) 646–696.
[20] K. Iwata, S. Suzuki, R. Kai, R. Kurose, Direct numerical simulation of detonation–turbulence interaction in hydrogen/oxygen/argon mixtures with a detailed chemistry, Phys. Fluids 35 (2023) 046107.
[21] S. Suzuki, K. Iwata, R. Kai, R. Kurose, A DNS study of detonation in H2/O2 mixture with variable-intensity turbulences, Proc. Combust. Inst. 40 (2024) 105337.
[22] R. Borghi, On the Structure and Morphology of Turbulent Premixed Flames, Springer US, Boston, MA, 1985, pp. 117–138.
[23] N. Peters, Turbulent Combustion, Cambridge University Press, Cambridge, UK, 2000, pp. 78–85.
[24] S. Kao, J. Ziegler, N. Bitter, B. Schmidt, J. Lawson, J.E. Shepherd, Numerical tools for shock and detonation wave modeling, 2024.
[25] A. Baumgart, G. Beardsell, G. Blanquart, Analytical closure to the spatially-filtered Euler equations for shock-dominated flows, J. Comput. Phys. 477 (2023) 111946.
[26] S.B. Pope, Turbulent Flows, Cambridge University Press, Cambridge, UK, 2000, pp. 185–187.